

HS 582: Energy Economics

Course Code	HS 582
Course Name	Energy Economics
L-T-P-C	3-0-0-3
Intended for	B. Tech./M.S./Ph.D.
Distribution	Elective
Approval	10th Senate (Jun 2016)

Course Contents

- **Orientation** [3 Lectures]
 - The rejuvenation of energy economics- with OPEC and the oil price shocks in 1970s and with climate change debates in 1990s
 - Recapitulation of some basic concepts: behavior of consumer, producer, prosumer (production by consumer), elasticity, growth rate, resource rent, rate of discount, net and present value, internal rate of return, energy intensity
- **An introduction to energy resource** [3 Lectures]
 - Classification of energy resource- depletable and non-depletable, primary and secondary, commercial and non-commercial
 - Units, conversion factors and aggregations of energy flow
 - Energy accounting framework-introduction to Energy Balance Statistics with example from India;
- **Basics of energy demand** [15 Lectures]
 - Evolution of energy demand analysis;
 - Economic foundations of energy demand consumer demand for energy, producer demand (input demand) for energy;
 - Introduction to analytical frameworks- accounting approach (decomposition analysis), econometric approach and techno-economic approach;
 - Energy demand management;
 - Rebound effect
- **Basics of energy supply** [15 Lectures]
 - Depletable primary energy resource- economics of exploration, optimal extraction rule, investment decision, resource production- coal, oil, natural gas; relation between discovery and production, depletion dimension;
 - Economics of secondary energy supply (electricity);
 - Economics of renewable energy supply- growth curve and rate of exploitation; drivers of renewable energy, cost features, support mechanism (feed-in-tariff, competitive bidding process, renewable obligations)
 - New economic principle when conventional consumer-producer divide is blurred
- **Energy access** [3 Lectures]
 - Energy use ladder;
 - Indicators of energy poverty; Affordability
 - Energy poverty and/or environmental protection A critical analysis
- **Energy Security** [3 Lectures]
 - Indicators of energy security- dependence, concentration and diversity of supply; optimal level of energy dependence; Geopolitics
 - Policies to enhance energy security import restriction and diversification, diversification of fuel mix, energy efficiency improvement;
 - Trade-off between energy security and climate change mitigation

Course reading

1. Stevens, P., **An Introduction to Energy Economics**, In Stevens, P. (ed.) The Economics of Energy, Vol. 1, Edward Elgar, 2000.
2. Bhattacharyya, Subhes. C., **Energy Economics: Concepts, Issues, Markets and Governance**, Springer, 2011. (Selected chapters)
3. Hartwick, J. M, and Olewiler, N. D., **The Economics of Natural Resource Use**, Harper and Row Publishers, 1986.
4. GEA, 2012: Global Energy Assessment - Toward a Sustainable Future, Cambridge University Press.